

M002 · Redstone & Levers

Topic: Redstone Levers · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

The maze has **redstone levers**. The agent detects redstone, flips a lever with `interact` to open the door, then walks through to clear the maze.

1 · The API

Code	What it does
<code>agent.detect(REDSTONE, FORWARD)</code>	True if redstone is ahead
<code>agent.interact(...)</code>	flip the lever in front
<code>agent.move(FORWARD, 1)</code>	step one block
<code>agent.turn(RIGHT_TURN)</code>	turn right
<code>player.on_chat("go", run)</code>	run code on chat word

2 · Predict 🧠

```
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

The agent reaches a lever and flips it. What happens to the door?


```
agent.interact(...)
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
```

There is no lever in front. What happens?

3 · Trace the Code 🔍

```
while not agent.detect(REDSTONE, FORWARD):  
    agent.move(FORWARD, 1)  
agent.interact(...)  
agent.turn(RIGHT_TURN)  
agent.move(FORWARD, 1)  
agent.move(FORWARD, 1)
```

Describe what the agent does, step by step, in plain English.

4 · Spot and Fix the Bugs 🐛

Write the fixed code, then say why.

Bug A — Flip the lever **first**, then walk through the open door. The lever is at the first wall.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

Fixed code:

Why was it wrong?

Bug B — Walk **up to** the lever, then flip it **once**. This flips on every step.

```
while not agent.detect(REDSTONE, FORWARD):
    agent.move(FORWARD, 1)
    agent.interact(...)
```

Fixed code:

Why was it wrong?

Bug C — After flipping, the agent must **turn around** and go back through the door. The door is behind it.

```
agent.interact(...)
agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong?

5 · Write the Code 🖋️

Somewhere ahead is a lever. Write code that walks to it, flips it, and walks through the open door.

```
def run():
    # your code here

player.on_chat("go", run)
```

6 · Show Your Work 📷 🎥

Open your homework world. The maze has a **lever** that opens a **closed door**. Write code that:

1. moves the agent to the lever,
2. flips it with `interact` ,
3. walks through the open door,
4. reaches the goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.